

University of Ulm
Faculty of Mathematics and Economics

Machine Learning and Decision Making

First Topic

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Problem Statement

A Data pertaining to job seekers in India for the year (2021). Four files are named as per the number of records they contain while the fifth file is named as test.xls. The variables are self explanatory. However, the entries will need some cleaning up.

Special notes: Some of the salary figures are in Rs while some others are in different currencies. Consider only those which are in Rs

Lac indicates 100000

If you want to incorporate any other data (at the city level for control, you can do that. Mention your data source)

Question: Ask three interesting questions from the data and answer them. Question 1 must be solved using Clustering; Question 2 using Linear Regression and Question 3 using Logistic Regression.

Marks:

Data preparation and cleaning: 10 marks

Q1 (Clustering): 10 marks

Q2(Linear Regression): 10 marks

Q3 (Logistic Regression): 10 marks

Overall recommendation and findings based on the model: 10 marks

Data preparation and cleaning

```
> library(readxl)
> library(dplyr)
> library(stringr)
> library(car)
> library(caret)
> library(tidyverse)
> library(lubridate)
> library(ggplot2)
> library(purrr)
> library(tidyr)
> xlsx_path <- "C:/Users/USER/OneDrive/Документы/dmml/Job Seekers.xlsx"
> Job <- read_excel(xlsx_path) %>% as.data.frame() (1)
> #leaving only Rs.
> Job <- Job %>%
+   filter(str_detect(as.character(Salary), regex("(^|\\W) (rs)\\.\\.?.\\s*",
ignore_case = TRUE))) (2)
> #get a number from a row
> Job$Salary_lacs <- as.numeric(stringr::str_extract(Job$Salary,
"\\d+(?:\\.\\d+)?")) (3)
```

```

> nrow(Job)
[1] 179541
> View(Job)
> to_exp_years_int <- function(x) {
+   s <- tolower(as.character(x))
+
+   #extract the integers that come before "year(s)" and "month(s)"
+   yrs_raw <- stringr::str_extract(s, "\\b(\\d+)\\s*(?:year\\(s\\)|years?)\\b")
+   mos_raw <- stringr::str_extract(s,
+   "\\b(\\d+)\\s*(?:month\\(s\\)|months?)\\b")
+
+   yrs_num <- suppressWarnings(as.numeric(stringr::str_extract(yrs_raw,
+   "\\d+")))
+   mos_num <- suppressWarnings(as.numeric(stringr::str_extract(mos_raw,
+   "\\d+")))
+
+   #count them, treating missing values as 0, but if both are missing – set NA
+   y <- ifelse(is.na(yrs_num), 0, yrs_num)
+   m <- ifelse(is.na(mos_num), 0, mos_num)
+   res <- y + m / 12
+   res[is.na(yrs_num) & is.na(mos_num)] <- NA_real_
+   res
+ }
> Job$Experience_years <- to_exp_years_int(Job$Work_Experience)   (4)
> #fixed reference date
> ref_date <- as.Date("2025-08-31")
> #parse the DOB and compute the completed years as of the reference date
> Job$DOB_parsed <- as.Date(sub(" UTC$", "", as.character(Job$DOB)), format =
"%Y-%m-%d")
> age_num <- lubridate::time_length(lubridate::interval(Job$DOB_parsed,
ref_date), "years")
> #check for future dates
> age_num[Job$DOB_parsed > ref_date] <- NA_real_
> Job$Age_years <- floor(age_num)   (5)
> nrow(Job)
[1] 179541
> View(Job)
> Job_old <- Job
> Job$Course_Highest <- Job$`Course(Highest_Education)`
> Job <- subset(Job, select = -c(DOB, Gender, Salary, Work_Experience,
DOB_parsed, `Course(Highest_Education)`))   (5)
> Job <- Job %>% mutate(Functional_Area = str_trim(str_extract(Functional_Area,
"^[^,]+")) )

```

```

> Job <- Job %>% mutate(Area_of_Specialization =
str_trim(str_extract(Area_of_Specialization, "^[^,]+")) )
> Job <- Job %>% mutate(Industry =
str_trim(str_extract(Industry, "^[^,]+")) ) (7)

```

- 1) First, the data were imported from the Excel file Job Seekers.xlsx into a tabular format for further processing.
- 2) In line with the assignment requirement, only observations where the salary is reported in “Rs” were retained; other currencies were filtered out.
- 3) From the salary text field, the first numeric value was extracted into Salary_lacs and interpreted as lacs per year (lac = 100,000 Rs).
- 4) The Work_Experience field contains text of the form “X years Y months.” We extracted the integers for years and months and computed total experience in years as years + months/12. If both years and months were missing, the value was marked as missing (NA).
- 5) Dates of birth were parsed and full years of age were computed as of 31.08.2025. If a date of birth was (erroneously) later than this reference date, age was set to NA.
- 6) For modeling convenience, the field Course(Highest_Education) was duplicated under the simpler name Course_Highest; the original and other technical fields not required for modeling were removed. A backup copy of the data frame was saved before these changes.
- 7) In Functional_Area, Area_of_Specialization, and Industry multiple tags are often listed comma-separated. To prevent an explosion in the number of factor levels and to keep categories mutually exclusive, only the first tag (up to the first comma) was retained and whitespace trimmed. This is a deliberate simplification with a minor loss of information about secondary tags.

Q1 (Clustering)

#Q1

#"Can distinct clusters of candidates be identified among job seekers based on their profiles (age, experience, expected salary)?"

```

> Job1 <- Job %>%
+ dplyr::select(Salary_lacs, Experience_years, Age_years) %>%
+ na.omit()
> nrow(Job)
[1] 179541
> nrow(Job1)
[1] 179540

```

We restricted the analysis to the three variables directly tied to the research question—**Salary_lacs**, **Experience_years**, and **Age_years**—and dropped rows with missing values in any of these fields (listwise deletion). This reduced the sample by 1 observation (from 179,541 to 179,540), ensuring that distances and principal components are computed on complete cases.

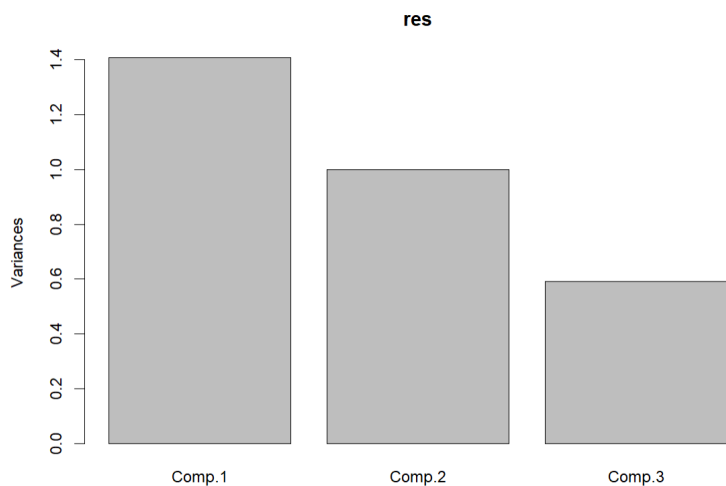
```

> # 1) baseline scale for exploratory PCA
> scaled_data <- scale(Job1[, c("Salary_lacs", "Experience_years", "Age_years")])

```

All three variables were standardized (z-scores) prior to PCA and clustering. Standardization is essential because the variables are on different scales (years vs. lacs). Without scaling, the variable with the largest variance would dominate Euclidean distances and hence both PCA and k-means.

```
> res <- princomp(scaled_data)
> plot(res)
```



```
> summary(res)
```

Importance of components:

	Comp.1	Comp.2	Comp.3
Standard deviation	1.1867802	0.9998188	0.7693494
Proportion of Variance	0.4694851	0.3332144	0.1973006
Cumulative Proportion	0.4694851	0.8026994	1.0000000

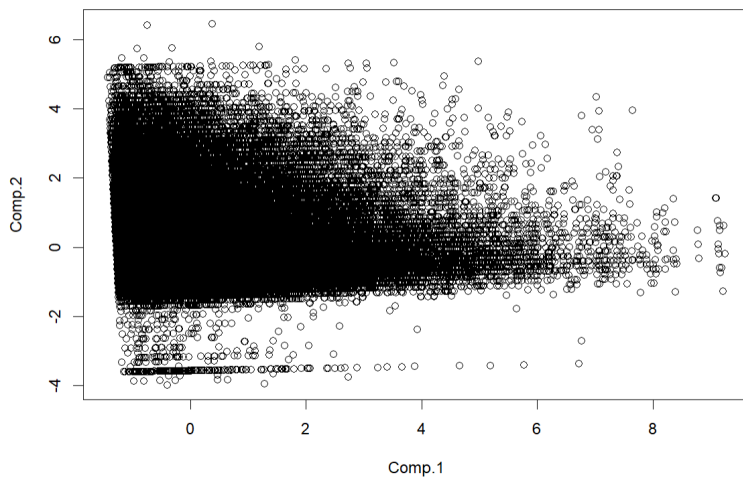
The summary reports the standard deviation of each PC and the variance share they explain:

- PC1: SD = 1.187 → **46.95%** of total variance
- PC2: SD = 1.000 → **33.32%** of total variance
- PC3: SD = 0.769 → **19.73%** of total variance

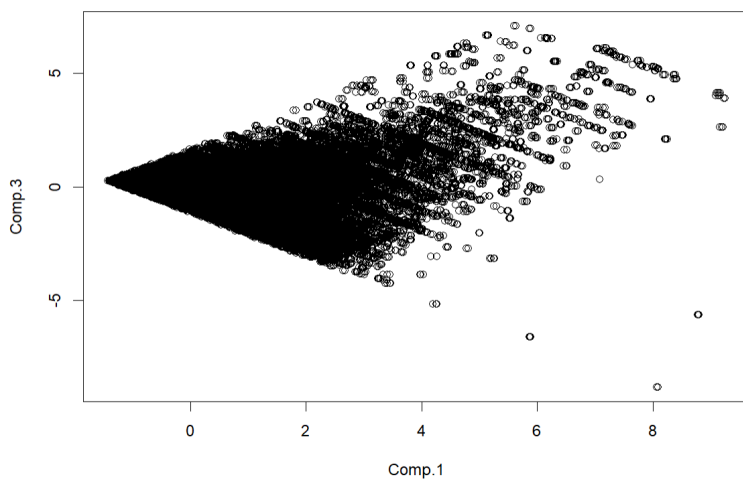
Thus, **PC1+PC2 explain ~80.27%** of the variability.

```
> #2D/3D views:
```

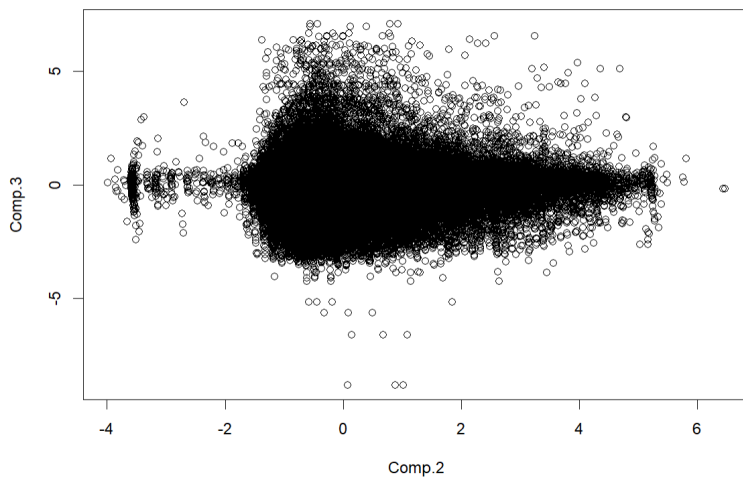
```
> plot(res$scores[,c(1,2)])
```



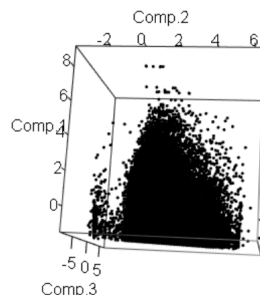
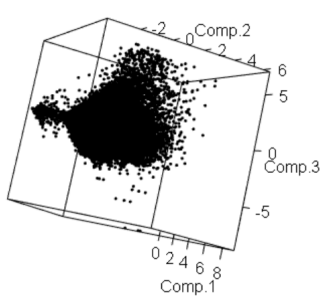
```
> plot(res$scores[,c(1,3)])
```



```
> plot(res$scores[,c(2,3)])
```



```
> library(rgl)
> plot3d(res$scores[,c(1,2,3)])
```



```
> # 2) preparing data for clustering (harmonizing columns)
```

```
> ClustData <- Job1 %>%
+   transmute(
+     .id = dplyr::row_number(),
+     Experience_years = as.numeric(Experience_years),
+     Age_years       = as.numeric(Age_years),
+     Salary_lac      = as.numeric(Salary_lacs)
+   ) %>%
+   drop_na()
```

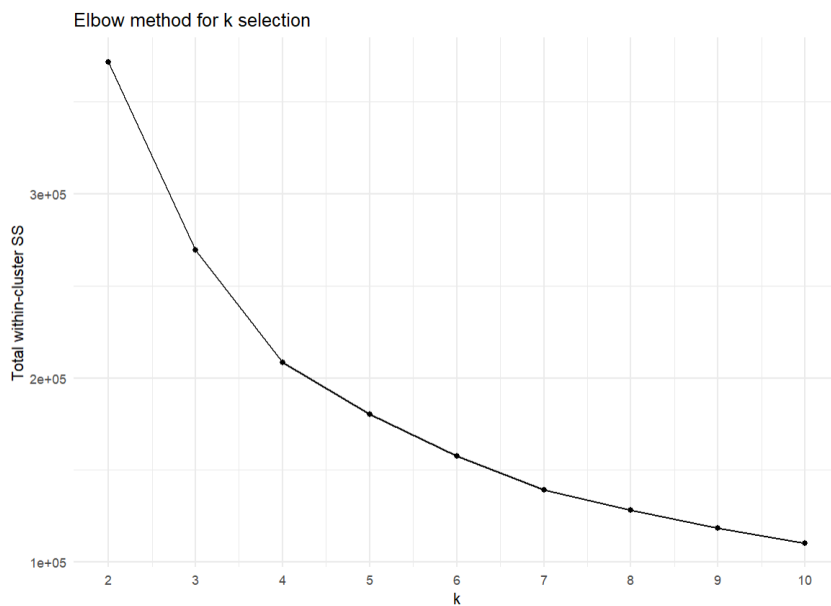
For clustering, I first built a working table that keeps only the variables relevant to the question—**Experience_years**, **Age_years**, and **Salary_lacs**—explicitly cast them to numeric, attached a row index for merging results back, and removed any rows with missing values so distances and principal components are computed on complete cases.

```
> X <- scale(ClustData[, c("Experience_years", "Age_years", "Salary_lac")])
```

All three features were standardized to z-scores (mean 0, variance 1) before running PCA and k-means. Standardization is crucial because the inputs are on different scales (years vs. lacs); without it, the variable with the largest dispersion would dominate Euclidean distances and bias the clusters.

```
> # 3) elbow method for choosing k
```

```
> set.seed(123)
> k_grid <- 2:10
> wss <- purrr::map_dbl(k_grid, ~ kmeans(X, centers = .x, nstart = 25, iter.max =
100)$tot.withinss)
> elbow_df <- data.frame(k = k_grid, wss = wss)
> ggplot(elbow_df, aes(k, wss)) +
+   geom_line() +
+   geom_point() +
+   scale_x_continuous(breaks = k_grid) +
+   labs(title = "Elbow method for k selection", x = "k", y = "Total within-cluster
SS") +
+   theme_minimal()
```



To select the number of clusters, I applied the elbow method: k-means was run for $k=2, \dots, 10$ and for each k I recorded the total within-cluster sum of squares (WSS). I set a fixed random seed for reproducibility, used multiple random starts (`nstart = 25`) to avoid poor local minima, and allowed sufficient iterations (`iter.max = 100`) for convergence. Plotting WSS against k produced the characteristic elbow curve.

The curve drops steeply up to about $k=4$ and then flattens, indicating diminishing returns beyond four clusters. Accordingly, I adopt **$k=4$** as a parsimonious choice.

```
> # 4) k-means
> # the most noticeable bend is around  $k \approx 4$  (after 4, the decrease becomes
noticeably slower)
> set.seed(123)
> k <- 4
> km <- kmeans(X, centers = k, nstart = 50, iter.max = 200)
> # 5) return clusters to Job
> Job1$Cluster <- NA_integer_
> Job1$Cluster[ClustData$id] <- km$cluster
> Job1$Cluster <- factor(Job1$Cluster)
> # 6) dataframe for 2D visualization overview (experience-salary)
> P <- Job1 %>%
+   transmute(
+     Experience_years = as.numeric(Experience_years),
+     Salary_lac       = as.numeric(Salary_lacs),
+     Cluster          = factor(Cluster)
+   ) %>%
+   drop_na()
> ggplot(P, aes(x = Experience_years, y = Salary_lac, colour = Cluster)) +
+   geom_point(alpha = 0.7) +
+   labs(
+     title = "Candidate Clustering: Experience vs Salary (lacs)",
+     x = "Experience (years)",
+     y = "Salary (lacs)"
+   ) +
+   theme_minimal()
```




The scatterplot of Experience (years) versus Salary (lacs), colored by cluster, provides a first sanity check and a visual interpretation of the segmentation: points form four density regions that separate primarily along the salary axis and secondarily along experience. One group concentrates at very low experience and low pay (entry-level mass), a second expands to moderate experience but remains in the lower-pay band, a third spans a wider experience range with clearly higher salaries (mid/upper segment), and a small, sparse group captures extremes (very high salaries and/or unusually high experience).

The k-means partitioning was learned in a standardized three-dimensional space (Experience_years, Age_years, Salary_lacs), whereas the scatterplot shows only the Experience-Salary projection. Part of the separation occurs along Age_years and along linear combinations of all three features. In addition, the strong right-skew of salaries (driven by Cluster 2) stretches the vertical axis, making lower-salary regions appear more overlapped in the 2D view.

> # 7) cluster profiles (in convenient units)

```
> cluster_profiles <- Job1 %>%
+   filter(!is.na(Cluster)) %>%
+   group_by(Cluster) %>%
+   summarise(
+     n = dplyr::n(),
+     mean_experience = mean(Experience_years, na.rm = TRUE),
+     mean_age = mean(Age_years, na.rm = TRUE),
+     mean_salary_lac = mean(Salary_lacs, na.rm = TRUE),
+     median_salary_lac = median(Salary_lacs, na.rm = TRUE),
+     .groups = "drop"
+   )
> print(cluster_profiles)
```

A tibble: 4 × 6

	Cluster	n	mean_experience	mean_age	mean_salary_lac	median_salary_lac
	<fct>	<int>	<dbl>	<dbl>	<dbl>	<dbl>
1	1	31517	22.2	44.6	12.2	11
2	2	9436	20.0	45.8	41.2	36
3	3	29169	9.16	59.2	7.82	6.02
4	4	109418	7.74	43.5	7.03	5.64

Cluster 1 (~17.6%) – Highly experienced with moderate pay

Average experience ≈ 22.2 years; average age ≈ 44.6 ; median salary ≈ 11 lacs (mean ≈ 12.2). This segment appears to comprise senior candidates whose compensation is not commensurate with tenure—potentially reflecting region, industry, or role types with lower pay scales.

Cluster 2 (~5.3%) – High earners

Average experience ≈ 20.0 years; average age ≈ 45.8 ; median salary ≈ 36 lacs (mean ≈ 41.2 , indicating a long right tail). This group likely includes top roles, Tier-1 cities, or scarce skill profiles.

Cluster 3 (~16.2%) – Older candidates with low to moderate pay

Average experience ≈ 9.2 years; average age ≈ 59.2 ; median salary ≈ 6.0 lacs (mean ≈ 7.8). This may reflect late-career transitions or part-time arrangements.

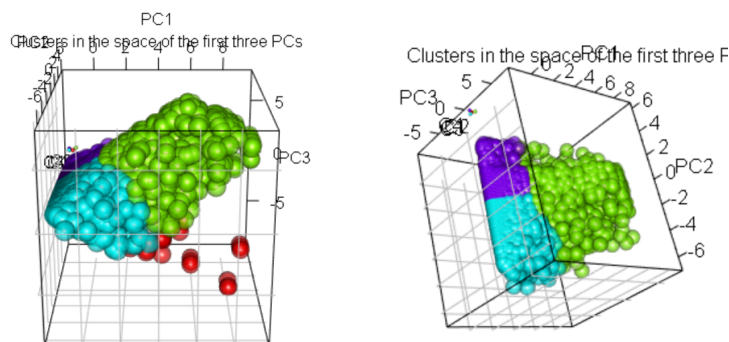
Cluster 4 (~60.9%) – Mass “mid-pool”

Average experience ≈ 7.7 years; average age ≈ 43.5 ; median salary ≈ 5.6 lacs (mean ≈ 7.0). This is the largest segment with moderate salary expectations.

After 15–20 years of experience, the population bifurcates: some candidates transition into high pay bands (Cluster 2), while others remain at moderate levels (Cluster 1). Salary does not increase monotonically with experience, and age alone does not guarantee higher compensation.

># 8) 3D visualization in the PC1-PC3 space

```
> pc <- prcomp(X, center = FALSE, scale. = FALSE)
> scores3 <- pc$x[, 1:3]
> cols <- rainbow(k)
> open3d()
wgl
  2
> plot3d(scores3[,1], scores3[,2], scores3[,3],
+       col = cols[km$cluster], size = 3, type = "s", alpha = 0.7,
+       xlab = "PC1", ylab = "PC2", zlab = "PC3")
> title3d(main = "Clusters in the space of the first three PCs")
> grid3d(c("x","y","z"))
> # centroids projected into the PC space
> centers_pc <- predict(pc, km$centers)
> spheres3d(centers_pc[,1], centers_pc[,2], centers_pc[,3],
+          col = cols, radius = 0.15)
> texts3d(centers_pc[,1], centers_pc[,2], centers_pc[,3],
+          texts = paste0("C", 1:k), adj = c(1.2, 1.2))
```



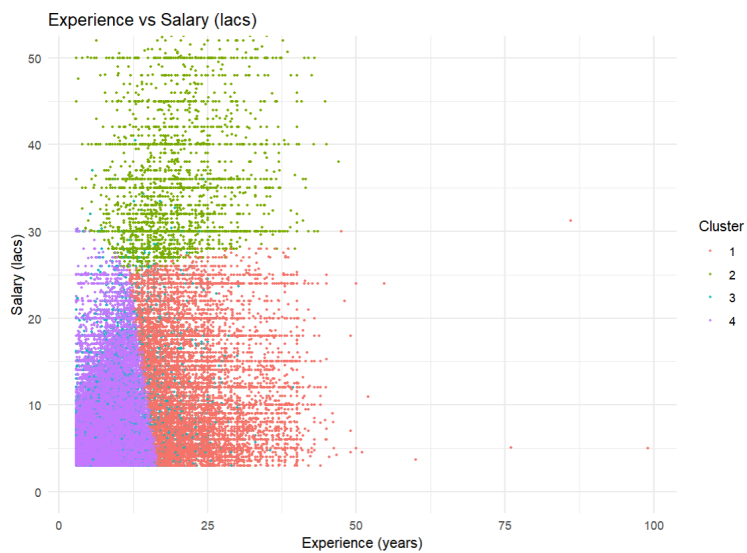
Answer to Q1 – “Can distinct clusters of candidates be identified based on age, experience, and expected salary?” – **Yes**.

We can observe four clear segments:

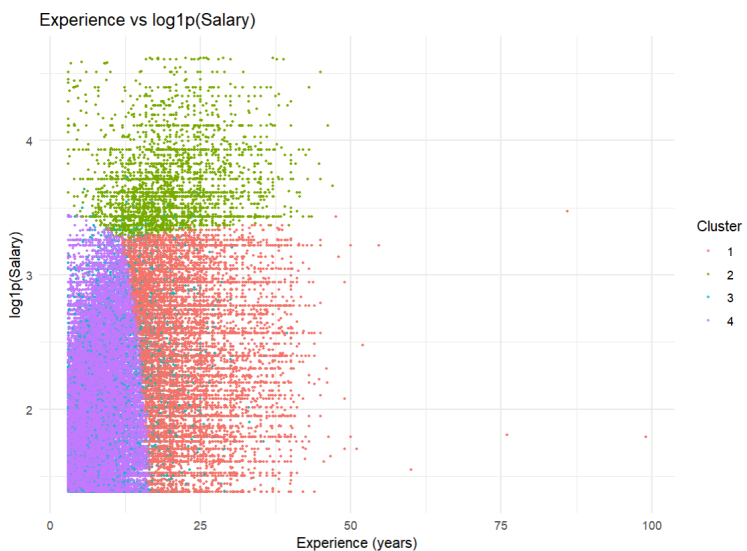
- **Highly experienced, moderate pay** (Seasoned professionals whose compensation appears restrained relative to their tenure—often tied to region, industry, or role type)
- **High earners** (A smaller, distinct group with strong experience profiles and substantially higher salary expectations, typical of top roles, premium locations, or scarce skills)
- **Older, low-to-moderate pay** (Later-career candidates whose salaries remain modest despite senior age; may reflect career switches, part-time roles, or market frictions)
- **Mass mid-pool** (The largest group, with moderate experience and moderate salary expectations)

After many years of experience the market **bifurcates**: some candidates transition into high-pay bands, while others remain at moderate levels. Compensation therefore does **not** rise in a simple linear fashion with experience, and age by itself does not guarantee higher pay.

```
> # dataframe for plots
> P <- Job1 %>%
+   select(Experience_years, Age_years, Salary_lacs, Cluster) %>%
+   filter(complete.cases(.)) %>%
+   mutate(Cluster = factor(Cluster))
> # trim very large salaries only for 2D visualization
> q99 <- quantile(P$Salary_lacs, 0.99, na.rm = TRUE)
> # experience vs salary (in lacs) with tail trimming
> ggplot(P, aes(Experience_years, Salary_lacs, colour = Cluster)) +
+   geom_point(alpha = 0.7, size = 0.7) +
+   coord_cartesian(ylim = c(0, q99)) +
+   labs(title = "Experience vs Salary (lacs)",
+        x = "Experience (years)",
+        y = "Salary (lacs)") +
+   theme_minimal()
```



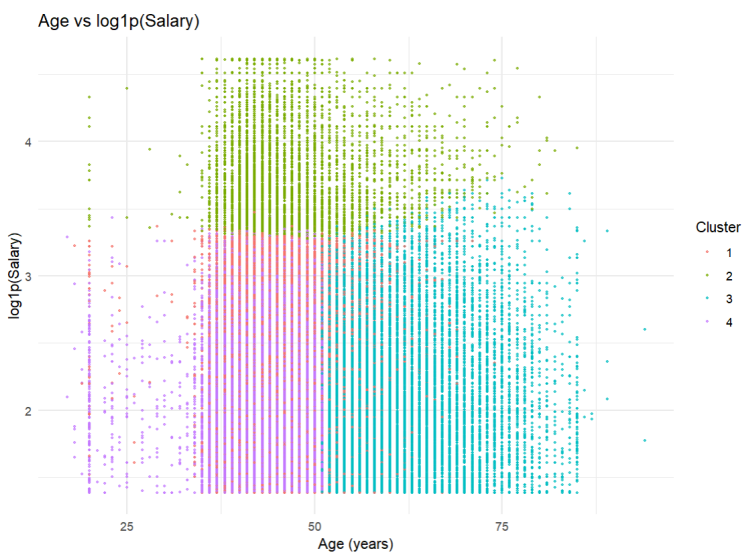
```
> # experience vs log1p(salary)
> ggplot(P, aes(Experience_years, log1p(Salary_lacs), colour = Cluster)) +
+   geom_point(alpha = 0.7, size = 0.7) +
+   labs(title = "Experience vs log1p(Salary)",
+         x = "Experience (years)",
+         y = "log1p(Salary)") +
+   theme_minimal()
```



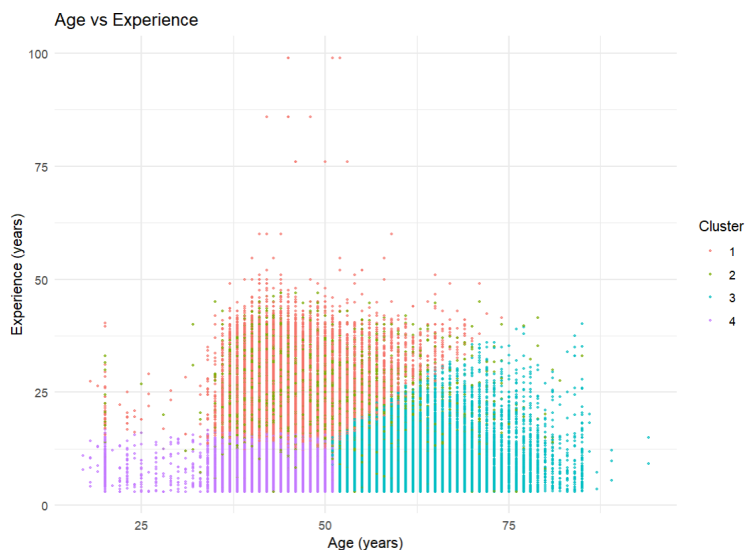
```
> # age vs salary (lacs) + trimming at the 99th percentile
> ggplot(P, aes(Age_years, Salary_lacs, colour = Cluster)) +
+   geom_point(alpha = 0.7, size = 0.7) +
+   coord_cartesian(ylim = c(0, q99)) +
+   labs(title = "Age vs Salary (lacs)",
+         x = "Age (years)",
+         y = "Salary (lacs)") +
+   theme_minimal()
```



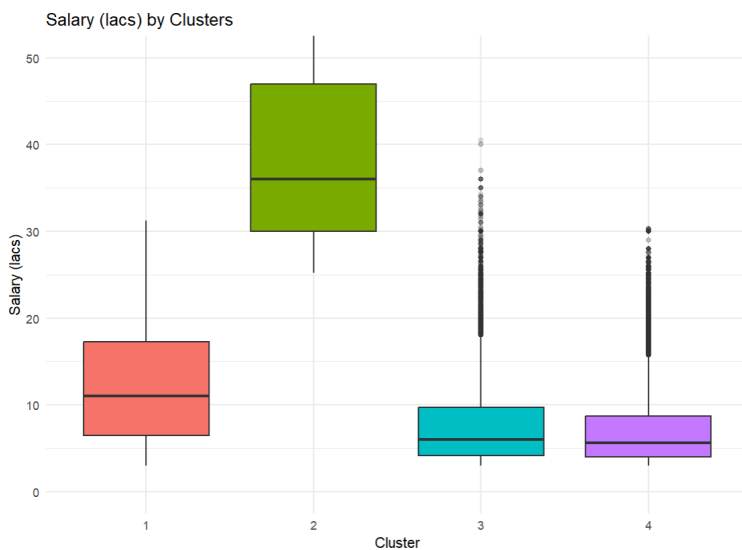
```
> # age vs log1p(salary)
> ggplot(P, aes(Age_years, log1p(Salary_lacs), colour = Cluster)) +
+   geom_point(alpha = 0.7, size = 0.7) +
+   labs(title = "Age vs log1p(Salary)",
+         x = "Age (years)",
+         y = "log1p(Salary)") +
+   theme_minimal()
```



```
> # age vs experience
> ggplot(P, aes(Age_years, Experience_years, colour = Cluster)) +
+   geom_point(alpha = 0.7, size = 0.7) +
+   labs(title = "Age vs Experience",
+         x = "Age (years)",
+         y = "Experience (years)") +
+   theme_minimal()
```



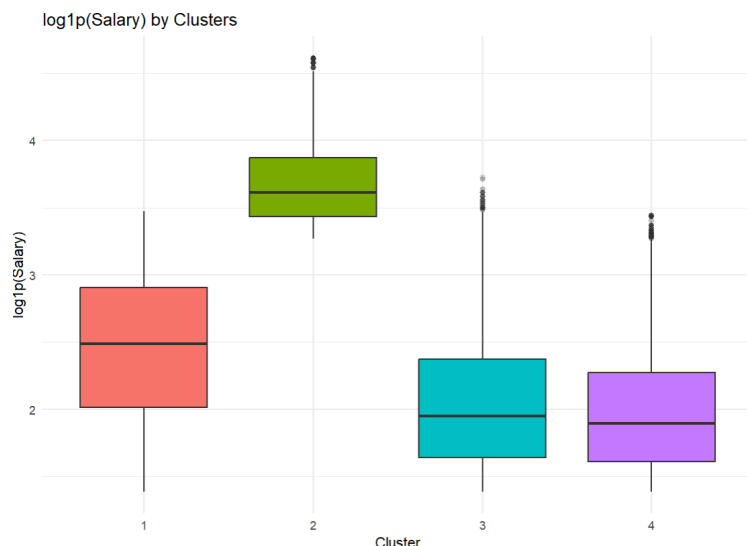
```
> # salary distribution across clusters: box plot (lacs, with tail trimming)
> ggplot(P, aes(Cluster, Salary_lacs, fill = Cluster)) +
+   geom_boxplot(outlier.alpha = 0.15) +
+   coord_cartesian(ylim = c(0, q99)) +
+   labs(title = "Salary (lacs) by Clusters",
+         x = "Cluster",
+         y = "Salary (lacs)") +
+   theme_minimal() +
+   theme(legend.position = "none")
```



The distributions separate clearly: one cluster forms a high-earning segment with a markedly higher median and the widest interquartile range; one cluster sits at a moderate salary level; and two clusters occupy the lower-salary regime with overlapping spreads (one slightly above the other).

```
> # salary distribution across clusters on the log1p scale (the differences
between centers are more visible)
> ggplot(P, aes(Cluster, log1p(Salary_lacs), fill = Cluster)) +
+   geom_boxplot(outlier.alpha = 0.15) +
+   labs(title = "log1p(Salary) by Clusters",
+         x = "Cluster",
+         y = "log1p(Salary)") +
+   theme_minimal() +
```

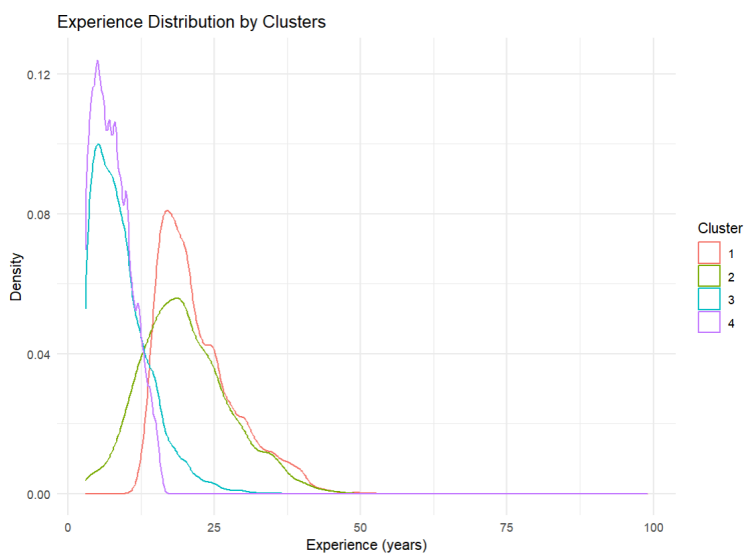
```
+ theme(legend.position = "none")
```



Using a log transform compresses the long right tail of salaries, so the central tendencies and spreads are easier to compare. One cluster clearly occupies the high-earning regime with a higher center and wider dispersion; one cluster sits at a moderate level; and two clusters overlap in the lower-salary range, with one slightly above the other. The transformation reduces the visual influence of extreme values without changing the rank ordering of clusters.

```
> # experience densities by clusters
```

```
> ggplot(P, aes(Experience_years, colour = Cluster)) +
+   geom_density(adjust = 1.2) +
+   labs(title = "Experience Distribution by Clusters",
+         x = "Experience (years)",
+         y = "Density") +
+   theme_minimal()
```



Each curve shows a kernel-density estimate of years of experience within a cluster (areas integrate to 1, so heights reflect shape, not group size). The high-earning cluster is clearly shifted to the right—its mass sits at higher experience. The “very experienced, moderate pay” cluster is also right-shifted. Two lower-pay clusters concentrate in the single-digit/low-teens range, with noticeable overlap between them. The broad overlap across ~5–15 years indicates that experience alone does not separate groups cleanly.

Q2(Linear Regression)

#Q2

Which factor influences salary more: additional years of experience or advancement to a higher job level?

```
> Job2 <- Job %>%
+   dplyr::select(Salary_lacs, Experience_years, Age_years,
+                 Course_Highest, Level, Industry) %>%
+   na.omit()
> nrow(Job)
[1] 179541
> nrow(Job2)
[1] 178011
> # 1) Baseline model (experience + age)
> modell <- lm(Salary_lacs ~ Experience_years + Age_years, data = Job2)
> summary(modell)
```

Call:

```
lm(formula = Salary_lacs ~ Experience_years + Age_years, data = Job2)
```

Residuals:

Min	1Q	Median	3Q	Max
-51.669	-3.916	-1.878	1.703	90.313

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	4.271483	0.135947	31.420	< 2e-16 ***
Experience_years	0.532589	0.002822	188.753	< 2e-16 ***
Age_years	-0.007311	0.002813	-2.599	0.00935 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.76 on 178008 degrees of freedom

Multiple R-squared: 0.1668, Adjusted R-squared: 0.1668

F-statistic: 1.782e+04 on 2 and 178008 DF, p-value: < 2.2e-16

```
> vif(modell)
```

Experience_years	Age_years
1.000062	1.000062

Model equation (in lacs):

Salary_lacs_hat = 4.27 + 0.533 × Experience_years - 0.0073 × Age_years

In this model the coefficient on Experience_years is 0.533 ($p < 0.001$): holding age constant, each additional year of experience is associated with an average increase of +0.533 lacs in expected salary. The coefficient on Age_years is -0.0073 ($p \approx 0.009$): conditional on experience, an extra year of age corresponds to a very small decrease of -0.0073 lacs. This statistical significance is likely driven by the very large sample size rather than by a practically meaningful effect.

Model fit is modest: $R^2 = 0.167$ (about 17% of salary variation explained), implying that much of the dispersion lies in factors not

included in this baseline. The residual standard error ≈ 8.76 laks indicates substantial heterogeneity in salary expectations. Multicollinearity is negligible (VIF ≈ 1.00 for both predictors), so the estimates are stable.

At this stage, experience has a much stronger and practically relevant association with salary than age (the age effect is economically negligible).

```
> # 2) Add education (Course(Highest Education))
> Job2$Course_Highest <- as.factor(Job2$Course_Highest)
> model2 <- lm(Salary_lacs ~ Experience_years + Age_years + Course_Highest, data = Job2)
> summary(model2)
```

Call:

```
lm(formula = Salary_lacs ~ Experience_years + Age_years + Course_Highest,
    data = Job2)
```

Residuals:

Min	1Q	Median	3Q	Max
-52.104	-3.917	-1.874	1.706	90.503

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	4.168935	0.170102	24.508	< 2e-16	***
Experience_years	0.532057	0.002823	188.504	< 2e-16	***
Age_years	-0.007247	0.002813	-2.576	0.00999	**
Course_HighestAviation	0.010630	0.409213	0.026	0.97928	
Course_HighestB.A.	0.114872	0.119551	0.961	0.33662	
Course_HighestB.Arch	-0.136329	0.468964	-0.291	0.77128	
Course_HighestB.Com	0.112581	0.117841	0.955	0.33940	
Course_HighestB.Ed	-0.198393	0.515521	-0.385	0.70036	
Course_HighestB.Pharm	-0.819906	0.506738	-1.618	0.10566	
Course_HighestB.Plan	0.019964	2.126865	0.009	0.99251	
Course_HighestB.Sc	0.223078	0.137037	1.628	0.10355	
Course_HighestBBA	-0.149956	0.240986	-0.622	0.53377	
Course_HighestBCA	-0.258030	0.186689	-1.382	0.16693	
Course_HighestBDS	0.755465	0.990951	0.762	0.44584	
Course_HighestBE/B.Tech	0.008630	0.122031	0.071	0.94362	
Course_HighestBGL	0.104127	3.098428	0.034	0.97319	
Course_HighestBHM	-0.209627	0.320751	-0.654	0.51340	
Course_HighestBL/LLB	-0.151316	0.229922	-0.658	0.51046	
Course_HighestBVSC	-1.219203	1.754849	-0.695	0.48721	
Course_HighestCA	0.222699	0.183023	1.217	0.22369	
Course_HighestCS	0.511944	0.456189	1.122	0.26177	
Course_HighestDiploma	0.072941	0.128744	0.567	0.57102	
Course_HighestDoctorate (Phd)	0.464895	0.388118	1.198	0.23099	
Course_HighestICWA	0.390204	0.329576	1.184	0.23643	
Course_HighestM.Arch	-0.238336	1.052059	-0.227	0.82078	
Course_HighestM.Com	-0.003387	0.183944	-0.018	0.98531	
Course_HighestM.Ed	0.720548	1.207629	0.597	0.55073	
Course_HighestM.Pharm	0.698541	0.895416	0.780	0.43532	
Course_HighestM.Phil	0.534364	0.426325	1.253	0.21005	
Course_HighestM.Sc	0.071224	0.165819	0.430	0.66754	

Course_HighestMA	-0.057330	0.171173	-0.335	0.73768
Course_HighestMBA/PGDM	0.122133	0.115755	1.055	0.29138
Course_HighestMBBS	-0.226414	0.466559	-0.485	0.62747
Course_HighestMCA/PGDCA	-0.138538	0.141140	-0.982	0.32632
Course_HighestMD/MS	0.315605	0.416042	0.759	0.44810
Course_HighestMDS	-0.319845	1.658538	-0.193	0.84708
Course_HighestME/M.Tech	0.305171	0.237368	1.286	0.19857
Course_HighestML/LLM	0.207947	0.691860	0.301	0.76375
Course_HighestOthers	0.587469	0.132602	4.430	9.41e-06 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.759 on 177972 degrees of freedom

Multiple R-squared: 0.1672, Adjusted R-squared: 0.167

F-statistic: 940 on 38 and 177972 DF, p-value: < 2.2e-16

> vif(model2)

	GVIF	Df	GVIF^(1/(2*Df))
Experience_years	1.000903	1	1.000452
Age_years	1.000256	1	1.000128
Course_Highest	1.001037	36	1.000014

># None of the course coefficients are statistically significant (all p-values > 0.1) → formally, education does not provide substantial additional information after accounting for experience and age.

We extend the baseline to

$\text{Salary_lacs} = \beta_0 + \beta_1 \cdot \text{Experience_years} + \beta_2 \cdot \text{Age_years} + \sum \gamma_i \cdot \text{Course_Highest}_i + \varepsilon$,
where Course_Highest enters as a set of dummy variables (one category is the reference).

The coefficients on the core covariates remain essentially unchanged:

* Experience_years = 0.532 (p < 0.001): holding age and education fixed, each extra year of experience is associated with +0.532 lakh higher salary.
* Age_years = -0.00725 (p ≈ 0.010): conditional on experience and education, age has a very small negative association (economically negligible).

Almost all individual Course_Highest categories are not statistically significant (most p > 0.10). The only notable coefficient is a positive effect for the miscellaneous "Others" bucket (p < 0.001). Because "Others" aggregates heterogeneous programs, it should not be over-interpreted as a specific educational premium; it is likely capturing residual mix/encoding effects rather than a clean causal signal.

Fit is virtually unchanged relative to the baseline:

* R^2 = 0.1672 (vs. 0.1668 before);
* Residual SE ≈ 8.76 lakh (unchanged to two decimals).

This indicates that education adds little explanatory power once experience (and age) are in the model.

Generalized VIFs implying no collinearity concerns and stable estimates. After controlling for experience and age, formal education categories do not materially improve salary prediction in this dataset. The economically meaningful driver remains years of experience.

```

> # 3) Adding job level (Level)
> Job2$Level <- as.factor(Job2$Level)
> model3 <- lm(Salary_lacs ~ Experience_years + Age_years + Level, data = Job2)
> summary(model3)

Call:
lm(formula = Salary_lacs ~ Experience_years + Age_years + Level,
    data = Job2)

Residuals:
    Min       1Q   Median       3Q      Max
-51.875  -3.917  -1.873   1.703   90.434

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)      5.087946   0.299627  16.981  < 2e-16
***
Experience_years    0.531182   0.002825 188.058  < 2e-16
***
Age_years         -0.006480   0.002814  -2.303 0.021274 *
LevelEntry Level/Fresher -1.012162   0.271730  -3.725 0.000195
***
LevelJunior - Executive, Assistant -1.005836   0.270159  -3.723 0.000197
***
LevelMiddle - Manager, Assistant Manager -0.740623   0.269710  -2.746 0.006033 **
Levelnull         -1.957875   0.726645  -2.694 0.007052 **
LevelOthers       -1.344919   0.398503  -3.375 0.000739
***
LevelPart time    -1.018851   0.306040  -3.329 0.000871
***
LevelSenior -Sr. Manager, General Manager -0.508465   0.274608  -1.852 0.064085 .
LevelTop - Directors, VP      0.031135   0.319961   0.097 0.922482
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.757 on 178000 degrees of freedom
Multiple R-squared:  0.1673, Adjusted R-squared:  0.1673
F-statistic: 3577 on 10 and 178000 DF, p-value: < 2.2e-16

> vif(model3)

                GVIF Df GVIF^(1/(2*Df))
Experience_years 1.002732  1      1.001365
Age_years       1.001116  1      1.000558

```

```

Level          1.003744  8          1.000234
># Transitioning to a managerial level results in a salary increase equivalent to
approximately 3-5 additional years of experience (depending on the specific
level).
> beta_exp <- coef(model3) ["Experience_years"]
> lev_eff  <- coef(model3) [grep("^Level", names(coef(model3)))]
> years_equiv <- sort(lev_eff / beta_exp)  # "years equivalent"
> round(years_equiv, 2)
> round(years_equiv, 2)

```

Level	LevelNull	LevelPart time	LevelJunior - Executive, Assistant	LevelMiddle - Manager, Assistant Manager	LevelSenior -Sr. Manager, General Manager	LevelEntry Level/Fresher	LevelOthers	LevelTop - Directors, VP
	-3.69	-1.92	-1.89	-1.39	-0.96	-2.53	-1.91	0.06

We estimated a linear regression of salary (in lacs) on experience, age, and job level. The baseline level is "Chief Officer - CEO/CMO/COO/CIO/President." The fitted model is:

```

Salary_lacs = 5.0879 + 0.5312·Experience_years - 0.00648·Age_years
- 1.0122·I[Level = Entry Level/Fresher]
- 1.0058·I[Level = Junior - Executive, Assistant]
- 0.7406·I[Level = Middle - Manager, Assistant Manager]
- 1.9579·I[Level = null]
- 1.3449·I[Level = Others]
- 1.0189·I[Level = Part time]
- 0.5085·I[Level = Senior - Sr. Manager, General Manager]
+ 0.0311·I[Level = Top - Directors, VP]

```

where I[.] are indicator (dummy) variables and coefficients are in lacs.

Statistical significance:

```

* Experience_years: p < 0.001
* Age_years: p ≈ 0.021
* Significant levels (p < 0.01): Entry, Junior, Middle, Others, Part time; Senior
is borderline (p ≈ 0.064); Top - Directors, VP is not significant.

```

Model fit: $R^2 \approx 0.167$, Residual SE ≈ 8.76 lacs. Multicollinearity check: VIF ≈ 1 for all predictors.

Given the experience slope is 0.5312 lacs per year, level differences can be expressed as an "equivalent years of experience" (holding age and other factors fixed):

```

* From Chief to Entry/Junior/Part time: about -1.9 years.
* From Chief to Middle: ≈ -1.39 years.
* From Chief to Senior: ≈ -0.96 years.
* From Chief to Top (Directors/VP): ≈ +0.06 years (small and not significant).
Experience exerts a stronger and more predictable influence on pay, while level
effects are uneven and category-dependent.

```

Answer to Q2 – “Which influences salary more: extra years of experience or moving up one job level?” – **extra years of experience have a stronger and more stable effect than a single “step” up in job level.**

Explanation (Model 3):

Baseline experience effect:

$\beta_{\text{exp}} = 0.531182$ lacs per additional year.

We compare adjacent job levels by taking the difference of their coefficients and convert that difference into an “experience-equivalent” by dividing by β_{exp} .

- **Junior → Middle**

Difference: $(-0.740623) - (-1.005836) = +0.265213$ lacs

Experience-equivalent: $0.265213 / 0.531182 \approx 0.50$ years

- **Middle → Senior**

Difference: $(-0.508465) - (-0.740623) = +0.232158$ lacs

Experience-equivalent: $0.232158 / 0.531182 \approx 0.44$ years

- **Senior → Top** (*interpret with caution: the Top coefficient vs. the base is not statistically significant*)

Difference: $(+0.031135) - (-0.508465) = +0.539600$ lacs

Experience-equivalent: $0.539600 / 0.531182 \approx 1.02$ years

- **Entry → Junior**

Difference: $(-1.005836) - (-1.012162) = +0.006326$ lacs

Experience-equivalent: $0.006326 / 0.531182 \approx 0.01$ years

A typical one-level promotion yields a gain equivalent to roughly 0.4–1.0 years of additional experience. By contrast, each extra year of experience consistently adds ≈ 0.531 lacs regardless of level. Therefore, accumulating relevant experience is a more powerful and predictable driver of salary than title changes, whose effects are smaller and less consistent and depend on the specific transition.

> # 4) Add industry (Industry)

> Job2\$Industry <- as.factor(Job2\$Industry)

> model4 <- lm(Salary_lacs ~ Experience_years + Age_years + Industry, data = Job2)

> summary(model4)

Call:

```
lm(formula = Salary_lacs ~ Experience_years + Age_years + Industry,
    data = Job2)
```

Residuals:

Min	1Q	Median	3Q	Max
-51.808	-3.926	-1.870	1.700	90.408

Coefficients:

	Estimate	Std. Error	t
value Pr(> t)			
(Intercept)	4.354280	0.153676	
28.334 < 2e-16 ***			
Experience_years	0.531760	0.002823	
188.336 < 2e-16 ***			
Age_years	-0.007076	0.002813	
-2.515 0.01189 *			
IndustryAdvertising/PR/Event Management	0.106955	0.174890	
0.612 0.54083			
IndustryAgriculture/Forestry/Fishing	0.391140	0.370028	
1.057 0.29049			
IndustryApparel/Garments	-0.240886	0.186423	
-1.292 0.19631			
IndustryArchitecture/Interior Design	-0.994770	0.255572	
-3.892 9.93e-05 ***			
IndustryAutomobiles/Auto Component/Auto Ancillary	0.209909	0.160096	
1.311 0.18981			
IndustryBanking	-0.361028	0.121920	
-2.961 0.00306 **			
IndustryBiotechnology/Pharmaceutical/Medicine	-0.071028	0.161392	
-0.440 0.65987			
IndustryCatering/Food Services/Restaurant	-0.191299	0.362389	
-0.528 0.59758			
IndustryConstruction/Cement/Metal/Steel/Iron	-0.313011	0.162571	
-1.925 0.05418 .			
IndustryConsulting Services	-0.325471	0.202018	
-1.611 0.10716			
IndustryConsumer Durables/FMCG	0.127848	0.129564	
0.987 0.32376			
IndustryCRM/CallCentres/BPO/ITES/Med.Trans	-0.250987	0.114306	
-2.196 0.02811 *			
IndustryDefence	0.758097	0.273744	
2.769 0.00562 **			
IndustryEducational/Training	-0.256524	0.137416	
-1.867 0.06194 .			
IndustryEngineering/Projects	0.042828	0.113927	
0.376 0.70697			
IndustryEntertainment/Media	0.298685	0.142930	
2.090 0.03664 *			
IndustryExport/Import/Merchandising	-0.320226	0.193062	
-1.659 0.09718 .			

IndustryFashion/Modelling	-0.109442	0.757475
-0.144 0.88512		
IndustryFinancial Services/Stockbroking	-0.100017	0.143744
-0.696 0.48656		
IndustryGems & Jewellery	0.521553	0.690038
0.756 0.44975		
IndustryGovernment Sector	0.016495	0.278203
0.059 0.95272		
IndustryHealthcare	-0.099622	0.176190
-0.565 0.57179		
IndustryHotel/Travel/Tourism/Airlines/Hospitality	-0.067103	0.135030
-0.497 0.61922		
IndustryInsurance	0.079145	0.148772
0.532 0.59473		
IndustryInternet/Dot com/ISP	0.202958	0.342866
0.592 0.55389		
IndustryIT-Hardware/Networking	-0.198954	0.118049
-1.685 0.09192 .		
IndustryIT-Software	-0.104103	0.097827
-1.064 0.28726		
IndustryLaw/Legal Firms	-0.340148	0.229548
-1.482 0.13839		
IndustryLogistics/Freight Forward/Distribution/Courier	-0.403795	0.204231
-1.977 0.04803 *		
IndustryManufacturing/Industrial	0.076099	0.134521
0.566 0.57160		
IndustryMarine/Aviation/Military/Mining/Shipping	-0.359609	0.514504
-0.699 0.48459		
IndustryNGO/Social Services	-0.300754	0.338654
-0.888 0.37449		
IndustryOthers	-0.087340	0.100509
-0.869 0.38486		
IndustryPetrochemicals	0.438256	0.592528
0.740 0.45952		
IndustryPetroleum/Oil and Gas/Power	0.192280	0.253414
0.759 0.44800		
IndustryPrinting/Packaging	0.834048	0.288738
2.889 0.00387 **		
IndustryProjects/Infrastructure/Power/Energy	-0.027020	0.294671
-0.092 0.92694		
IndustryReal Estate	-0.316037	0.250299
-1.263 0.20672		

```

IndustryRecruitment/Placement Agencies      -0.209145    0.341917
-0.612    0.54075
IndustryResearch/Surveyor/MR                -0.522453    0.441028
-1.185    0.23617
IndustryRetailing                           0.040562    0.183396
0.221    0.82496
IndustrySecurity/Law Enforcement             0.293318    0.442083
0.663    0.50702
IndustrySoftware Services                   -0.478245    0.276891
-1.727    0.08414 .
IndustrySports/Recreation                   -0.721883    1.007391
-0.717    0.47363
IndustryTelecom                             -0.115183    0.145437
-0.792    0.42837
IndustryWarehousing                         -1.294350    0.505404
-2.561    0.01044 *

```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.758 on 177962 degrees of freedom

Multiple R-squared: 0.1673, Adjusted R-squared: 0.1671

F-statistic: 745.1 on 48 and 177962 DF, p-value: < 2.2e-16

> vif(model4)

```

              GVIF Df GVIF^(1/(2*Df))
Experience_years 1.001723  1          1.000861
Age_years       1.000544  1          1.000272
Industry        1.002150 46          1.000023

```

The specification explains a modest share of salary variation; unexplained dispersion remains sizeable. Diagnostic checks indicate no material multicollinearity among predictors. Tenure is the principal and most stable driver of pay, while age shows a slight negative association once tenure is held constant. Relative to the omitted baseline industry, some sectors are associated with higher salary levels (for example, printing/packaging, defence, and entertainment/media), whereas others are associated with lower levels (such as warehousing, architecture/interior design, banking, CRM/BPO/ITES, and logistics/courier). The IT-software sector does not display a distinct premium after controlling for experience and age. Overall, industry mainly produces modest upward or downward level shifts, while experience remains the most reliable determinant of salary.

> [# Compare models](#)

> AIC(model1, model2, model3, model4)

```

      df      AIC
model1  4 1277798

```



```
model2 40 1277799
model3 12 1277704
model4 50 1277783
```

Best among the considered specifications is **Model 3 (Experience + Age + Level)**, which has the lowest AIC (1,277,704).

Adding education (**Model 2**) does **not** improve fit relative to the baseline; AIC is essentially unchanged, so after accounting for experience and age, education contributes little additional explanatory power.

Adding industry (**Model 4**) yields a **worse** AIC than Model 3 (by ~79 points). Industry effects exist, but they do not outweigh the contribution of job level.

AIC differences on the order of ~80-95 points represent a **meaningful** advantage in favor of Model 3. For a good balance of accuracy and parsimony, use **Model 3**: experience and job level are the key predictors of salary; age has a weak negative effect. Education can be omitted, and industry may be included for interpretability rather than for model performance.

Additionally, we tested other specifications (expanded covariate sets, interaction terms, regularization, and stepwise selection). They did not outperform Model 3 by AIC. Given the noisy sample and many rare categories, models with more columns took considerably longer to compute—even with optimizations.

Q3 (Logistic Regression)

#Q3

Which factors are associated with a higher likelihood of earning a salary in the top quartile (top 25%)?

```
> library(dplyr)
> library(broom)
> library(pROC)
> library(caret)
> # Target: upper quartile
> q75 <- quantile(Job$Salary_lacs, 0.75, na.rm = TRUE)
> Job$HighPay <- as.integer(Job$Salary_lacs >= q75)
> # Prepare data without NA in model variables
> dat1 <- Job %>%
+   dplyr::select(HighPay, Experience_years, Age_years, Level, Industry,
+ Functional_Area) %>%
+   na.omit()
> nrow(dat1)
[1] 177509
```

We loaded the modeling packages (dplyr, broom, pROC, caret), defined the “high pay” threshold as the 75th percentile of Salary_lacs, and created a binary target HighPay (1 if salary ≥ that cutoff, 0 otherwise). We then assembled the modeling dataset with the predictors Experience_years, Age_years, Level, Industry, and Functional_Area, removed rows with missing values, and obtained a final sample of 177,509 observations. This prepared dataset will be used to fit a logistic regression that estimates the probability of being in the top-quartile of salaries.

```

> # Logistic model
> m1 <- glm(HighPay ~ Experience_years + Age_years + Level + Industry +
Functional_Area,
+           data = dat1, family = binomial())
> summary(m1)

Call:
glm(formula = HighPay ~ Experience_years + Age_years + Level +
    Industry + Functional_Area, family = binomial(), data = dat1)

Coefficients:
                                Estimate Std.
Error z value
(Intercept)                    -2.1530386
0.0846567 -25.433
Experience_years                  0.1138009
0.0008107 140.373
Age_years                       -0.0023398
0.0008064  -2.902
LevelEntry Level/Fresher        -0.2513646
0.0741511  -3.390
LevelJunior - Executive, Assistant -0.2469295
0.0733641  -3.366
LevelMiddle - Manager, Assistant Manager -0.1680906
0.0731051  -2.299
Levelnull                      -0.7434672
0.2369965  -3.137
LevelOthers                    -0.4417502
0.1169435  -3.777
LevelPart time                 -0.2327566
0.0844395  -2.756
LevelSenior -Sr. Manager, General Manager -0.1237213
0.0742870  -1.665
LevelTop - Directors, VP       -0.0450299
0.0863910  -0.521
IndustryAdvertising/PR/Event Management -0.0420992
0.0634165  -0.664
IndustryAgriculture/Forestry/Fishing -0.0130905
0.1094592  -0.120
IndustryApparel/Garments       -0.0632028
0.0595517  -1.061
IndustryArchitecture/Interior Design -0.1272417
0.1078335  -1.180
IndustryAutomobiles/Auto Component/Auto Ancillary 0.0450258
0.0512943   0.878
IndustryBanking                -0.0801572
0.0463320  -1.730
IndustryBiotechnology/Pharmaceutical/Medicine -0.0695419
0.0592185  -1.174
IndustryCatering/Food Services/Restaurant -0.0162792
0.1101825  -0.148
IndustryConstruction/Cement/Metal/Steel/Iron -0.1598313
0.0540205  -2.959

```

IndustryConsulting Services	-0.0798860
0.0634012 -1.260	
IndustryConsumer Durables/FMCG	0.0321275
0.0452706 0.710	
IndustryCRM/CallCentres/BPO/ITES/Med.Trans	-0.0004669
0.0425392 -0.011	
IndustryDefence	-0.0075997
0.0809099 -0.094	
IndustryEducational/Training	0.0215017
0.0495270 0.434	
IndustryEngineering/Projects	0.0874787
0.0445491 1.964	
IndustryEntertainment/Media	0.1018297
0.0495608 2.055	
IndustryExport/Import/Merchandising	-0.0335318
0.0641562 -0.523	
IndustryFashion/Modelling	0.3740541
0.2083971 1.795	
IndustryFinancial Services/Stockbroking	-0.0846536
0.0498511 -1.698	
IndustryGems & Jewellery	0.2308681
0.1868174 1.236	
IndustryGovernment Sector	-0.0871962
0.0835293 -1.044	
IndustryHealthcare	-0.0135132
0.0630616 -0.214	
IndustryHotel/Travel/Tourism/Airlines/Hospitality	0.0245137
0.0546764 0.448	
IndustryInsurance	0.0265767
0.0521160 0.510	
IndustryInternet/Dot com/ISP	0.0266572
0.1014163 0.263	
IndustryIT-Hardware/Networking	-0.0172710
0.0473049 -0.365	
IndustryIT-Software	-0.0103785
0.0428827 -0.242	
IndustryLaw/Legal Firms	-0.0207183
0.1087628 -0.190	
IndustryLogistics/Freight Forward/Distribution/Courier	-0.0191486
0.0640707 -0.299	
IndustryManufacturing/Industrial	-0.0062595
0.0448305 -0.140	
IndustryMarine/Aviation/Military/Mining/Shipping	-0.0157056
0.1486539 -0.106	
IndustryNGO/Social Services	-0.0722784
0.1032909 -0.700	
IndustryOthers	0.0271775
0.0376808 0.721	
IndustryPetrochemicals	0.0594939
0.1655353 0.359	
IndustryPetroleum/Oil and Gas/Power	-0.0159946
0.0812497 -0.197	
IndustryPrinting/Packaging	0.1370309
0.0860763 1.592	

IndustryProjects/Infrastructure/Power/Energy	0.1055416
0.0860287 1.227	
IndustryReal Estate	-0.0562610
0.0754384 -0.746	
IndustryRecruitment/Placement Agencies	0.0267994
0.1038897 0.258	
IndustryResearch/Surveyor/MR	-0.0383337
0.1296635 -0.296	
IndustryRetailing	0.0264887
0.0573833 0.462	
IndustrySecurity/Law Enforcement	-0.0369017
0.1398074 -0.264	
IndustrySoftware Services	-0.1327175
0.0864880 -1.535	
IndustrySports/Recreation	-0.3422167
0.3016182 -1.135	
IndustryTelecom	-0.0340621
0.0484328 -0.703	
IndustryWarehousing	-0.1289433
0.1531269 -0.842	
Functional_AreaAdministration	0.0219619
0.0386898 0.568	
Functional_AreaAdvertising/PR/Events	-0.0173773
0.0588305 -0.295	
Functional_AreaAnchoring/TV/Films/Production	-0.1583949
0.0871455 -1.818	
Functional_AreaArchitects/Interior Design/Naval Arch	-0.1190516
0.1150839 -1.034	
Functional_AreaBanks/Insurance/Financial Services	-0.0370423
0.0401942 -0.922	
Functional_AreaBio Tech/R&D/Scientist	0.0562698
0.0663990 0.847	
Functional_AreaBusiness Mgmt./Consultants/Freelancers	0.0500306
0.0637347 0.785	
Functional_AreaContent/Editors/Journalists	0.0509294
0.0667282 0.763	
Functional_AreaCorporate Planning/Consulting/Strategy	-0.0119049
0.0689276 -0.173	
Functional_AreaDirect Sales Agent/Insurance Agents	0.1057850
0.0757412 1.397	
Functional_AreaDoctors/Nurses/Medical Professional	0.0190270
0.0780688 0.244	
Functional_AreaElectronics	-0.0783265
0.0574459 -1.363	
Functional_AreaEngg. project Mgmt./site Engg./Civil Engg.	-0.0203498
0.0422404 -0.482	
Functional_AreaExport/Import	-0.0304138
0.0506156 -0.601	
Functional_AreaFinance	0.0580921
0.0535966 1.084	
Functional_AreaFresh Graduate-No Functional Area exp.	-0.0153470
0.0447357 -0.343	
Functional_AreaFront Office Staff/Secretarial/Computer Operator	-0.0292046
0.0471979 -0.619	

Functional_AreaHotel/Restaurant	0.0248155
0.0650915 0.381	
Functional_AreaHR/PM/IR/Training	-0.0236759
0.0476463 -0.497	
Functional_AreaIT/Telecom - Hardware	-0.0163990
0.0449457 -0.365	
Functional_AreaIT/Telecom - Software	0.0119350
0.0392870 0.304	
Functional_AreaLegal/Law	-0.0221002
0.1040438 -0.212	
Functional_AreaMarketing/MR/Media Planning	0.0316654
0.0392142 0.807	
Functional_AreaMaterials/Logistics/Supply Chain Mgmt/Procurement	0.0692612
0.0471541 1.469	
Functional_AreaOil and Gas	0.0749730
0.1093949 0.685	
Functional_AreaOperations/Customer Service/Telecalling/Backend	-0.0034412
0.0427898 -0.080	
Functional_AreaOthers	0.0056822
0.0401785 0.141	
Functional_AreaProduction/Manufacturing/Maintenance/Packaging	0.0197268
0.0478124 0.413	
Functional_AreaQuality/Process Control	-0.0665656
0.0657548 -1.012	
Functional_AreaSales/Business Development	-0.0275546
0.0357491 -0.771	
Functional_AreaSecurity	0.0914928
0.1477482 0.619	
Functional_AreaService Engineers	-0.0777860
0.0932942 -0.834	
Functional_AreaTeaching/Education	-0.1776367
0.0675374 -2.630	
Functional_AreaTicketing/Reservation/Travel Agent/Airlines	-0.1317619
0.0704458 -1.870	
Functional_AreaTop Management	-0.0614412
0.0913077 -0.673	
	Pr(> z)
(Intercept)	< 2e-16 ***
Experience_years	< 2e-16 ***
Age_years	0.003712 **
LevelEntry Level/Fresher	0.000699 ***
LevelJunior - Executive, Assistant	0.000763 ***
LevelMiddle - Manager, Assistant Manager	0.021488 *
Levelnull	0.001707 **
LevelOthers	0.000158 ***
LevelPart time	0.005843 **
LevelSenior -Sr. Manager, General Manager	0.095823 .
LevelTop - Directors, VP	0.602204
IndustryAdvertising/PR/Event Management	0.506784
IndustryAgriculture/Forestry/Fishing	0.904806
IndustryApparel/Garments	0.288549
IndustryArchitecture/Interior Design	0.238007
IndustryAutomobiles/Auto Component/Auto Ancillary	0.380056
IndustryBanking	0.083619 .

IndustryBiotechnology/Pharmaceutical/Medicine	0.240264
IndustryCatering/Food Services/Restaurant	0.882542
IndustryConstruction/Cement/Metal/Steel/Iron	0.003089 **
IndustryConsulting Services	0.207667
IndustryConsumer Durables/FMCG	0.477904
IndustryCRM/CallCentres/BPO/ITES/Med.Trans	0.991243
IndustryDefence	0.925167
IndustryEducational/Training	0.664186
IndustryEngineering/Projects	0.049571 *
IndustryEntertainment/Media	0.039914 *
IndustryExport/Import/Merchandising	0.601211
IndustryFashion/Modelling	0.072668 .
IndustryFinancial Services/Stockbroking	0.089484 .
IndustryGems & Jewellery	0.216535
IndustryGovernment Sector	0.296532
IndustryHealthcare	0.830324
IndustryHotel/Travel/Tourism/Airlines/Hospitality	0.653906
IndustryInsurance	0.610084
IndustryInternet/Dot com/ISP	0.792667
IndustryIT-Hardware/Networking	0.715037
IndustryIT-Software	0.808764
IndustryLaw/Legal Firms	0.848925
IndustryLogistics/Freight Forward/Distribution/Courier	0.765042
IndustryManufacturing/Industrial	0.888955
IndustryMarine/Aviation/Military/Mining/Shipping	0.915858
IndustryNGO/Social Services	0.484080
IndustryOthers	0.470751
IndustryPetrochemicals	0.719294
IndustryPetroleum/Oil and Gas/Power	0.843940
IndustryPrinting/Packaging	0.111391
IndustryProjects/Infrastructure/Power/Energy	0.219891
IndustryReal Estate	0.455796
IndustryRecruitment/Placement Agencies	0.796438
IndustryResearch/Surveyor/MR	0.767505
IndustryRetailing	0.644361
IndustrySecurity/Law Enforcement	0.791821
IndustrySoftware Services	0.124902
IndustrySports/Recreation	0.256542
IndustryTelecom	0.481878
IndustryWarehousing	0.399749
Functional_AreaAdministration	0.570279
Functional_AreaAdvertising/PR/Events	0.767705
Functional_AreaAnchoring/TV/Films/Production	0.069127 .
Functional_AreaArchitects/Interior Design/Naval Arch	0.300913
Functional_AreaBanks/Insurance/Financial Services	0.356746
Functional_AreaBio Tech/R&D/Scientist	0.396744
Functional_AreaBusiness Mgmt./Consultants/Freelancers	0.432463
Functional_AreaContent/Editors/Journalists	0.445323
Functional_AreaCorporate Planning/Consulting/Strategy	0.862874
Functional_AreaDirect Sales Agent/Insurance Agents	0.162515
Functional_AreaDoctors/Nurses/Medical Professional	0.807447
Functional_AreaElectronics	0.172730
Functional_AreaEngg. project Mgmt./site Engg./Civil Engg.	0.629975
Functional_AreaExport/Import	0.547922

Functional_AreaFinance	0.278419
Functional_AreaFresh Graduate-No Functional Area exp.	0.731553
Functional_AreaFront Office Staff/Secretarial/Computer Operator	0.536068
Functional_AreaHotel/Restaurant	0.703025
Functional_AreaHR/PM/IR/Training	0.619253
Functional_AreaIT/Telecom - Hardware	0.715213
Functional_AreaIT/Telecom - Software	0.761287
Functional_AreaLegal/Law	0.831785
Functional_AreaMarketing/MR/Media Planning	0.419380
Functional_AreaMaterials/Logistics/Supply Chain Mgmt/Procurement	0.141880
Functional_AreaOil and Gas	0.493128
Functional_AreaOperations/Customer Service/Telecalling/Backend	0.935902
Functional_AreaOthers	0.887535
Functional_AreaProduction/Manufacturing/Maintenance/Packaging	0.679909
Functional_AreaQuality/Process Control	0.311380
Functional_AreaSales/Business Development	0.440840
Functional_AreaSecurity	0.535753
Functional_AreaService Engineers	0.404410
Functional_AreaTeaching/Education	0.008534 **
Functional_AreaTicketing/Reservation/Travel Agent/Airlines	0.061428 .
Functional_AreaTop Management	0.501009

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 200673 on 177508 degrees of freedom
 Residual deviance: 176744 on 177417 degrees of freedom
 AIC: 176928

Number of Fisher Scoring iterations: 4

Logistic model m1 (target: being in the top 25% of salaries)

- Experience – strongest predictor. Coefficient = 0.1138 → OR = $\exp(0.1138) \approx 1.12$. Each additional year of experience increases the odds of being in the top quartile by about 12% ($p < 0.001$).
- Age – small but statistically significant negative effect. Coefficient = -0.00234 → OR ≈ 0.998 . Each additional year of age reduces the odds by $\sim 0.24\%$ ($p \approx 0.0037$), controlling for other factors.
- Job level (relative to the model's reference level):
 - Entry/Fresher: OR ≈ 0.78 ($\sim 22\%$ lower odds), $p < 0.001$
 - Junior: OR ≈ 0.78 , $p < 0.001$
 - Middle (Manager/Asst. Manager): OR ≈ 0.85 , $p \approx 0.021$
 - Part-time: OR ≈ 0.79 , $p \approx 0.006$
 - Senior (Sr. Manager/GM): OR ≈ 0.88 , $p \approx 0.096$ (borderline)
 - Top (Directors/VP): OR ≈ 0.96 , $p \approx 0.60$ (no meaningful difference)
 Interpretation: lower levels have clearly lower odds of being in the top quartile compared to the reference; for Top the difference is not detectable.
- Industries (relative to the reference industry):
 - Construction/Cement/Metal/Steel: OR ≈ 0.85 , $p \approx 0.003$ (lower odds)

- Engineering/Projects: OR ≈ 1.09 , $p \approx 0.050$ (higher odds)
- Entertainment/Media: OR ≈ 1.11 , $p \approx 0.040$ (higher odds)

Most other industries show no statistically significant differences.

- Functional areas (relative to the reference area):

- Teaching/Education: OR ≈ 0.84 , $p \approx 0.0085$ (lower odds)

Other functional areas are mostly non-significant or borderline.

- Model quality: AIC $\approx 176,928$. The reduction in deviance versus the null model is substantial; previously computed ROC gives AUC ≈ 0.77 , indicating moderate discrimination.

- Conclusion: After controlling for age, industry, and functional area, experience remains the primary driver of "high" pay ($\approx +12\%$ to the odds per additional year). Lower job levels substantially reduce the odds relative to the reference category. Selected industries and one functional area contribute additional—but smaller—effects.

```
> # Odds Ratios (OR) with 95% Confidence Intervals
```

```
> library(broom)
```

```
> z <- qnorm(0.975) # 1.96
```

```
> orl <- broom::tidy(ml, conf.int = FALSE) %>%
```

```
+ mutate(
```

```
+   OR = exp(estimate),
```

```
+   OR_lo = exp(estimate - z * std.error), # Wald 95% CI
```

```
+   OR_hi = exp(estimate + z * std.error)
```

```
+ ) %>%
```

```
+ arrange(p.value)
```

```
> orl %>% filter(term != "(Intercept)") %>% head(30)
```

```
# A tibble: 30 × 8
```

	term	estimate	std.error	statistic
p.value	OR OR_lo OR_hi			
	<chr>	<dbl>	<dbl>	<dbl>
<dbl>	<dbl>	<dbl>	<dbl>	
1	Experience_years	0.114	0.000811	140. 0
1.12	1.12 1.12			
2	LevelOthers	-0.442	0.117	-3.78
0.000158	0.643 0.511 0.809			
3	LevelEntry Level/Fresher	-0.251	0.0742	-3.39
0.000699	0.778 0.673 0.899			
4	LevelJunior - Executive, Assistant	-0.247	0.0734	-3.37
0.000763	0.781 0.677 0.902			
5	Levelnull	-0.743	0.237	-3.14
0.00171	0.475 0.299 0.757			
6	IndustryConstruction/Cement/Metal/Steel/Iron	-0.160	0.0540	-2.96
0.00309	0.852 0.767 0.947			
7	Age_years	-0.00234	0.000806	-2.90
0.00371	0.998 0.996 0.999			
8	LevelPart time	-0.233	0.0844	-2.76
0.00584	0.792 0.671 0.935			
9	Functional_AreaTeaching/Education	-0.178	0.0675	-2.63
0.00853	0.837 0.733 0.956			
10	LevelMiddle - Manager, Assistant Manager	-0.168	0.0731	-2.30
0.0215	0.845 0.732 0.975			
# i	20 more rows			
# i	Use `print(n = ...)` to see more rows			


```
> # Classification quality
> dat1$prob <- predict(m1, type = "response")
> roc1 <- pROC::roc(dat1$HighPay, dat1$prob); pROC::auc(roc1)
```

Setting levels: control = 0, case = 1

Setting direction: controls < cases

Area under the curve: 0.7707

Briefly: the model clearly shows that experience is the primary driver of landing in the top-25% of salaries; lower job levels, certain industries, and Teaching/Education are associated with lower odds; the classifier has moderate discrimination (AUC $\approx$ 0.77).

What the ORs and AUC indicate:

- Experience_years: OR $\approx$ 1.12 per additional year (95% CI $\sim$ 1.12-1.12; $p < 0.001$) – the largest and most stable effect. Each extra year increases the odds of being in the top quartile by about 12%.
- Age_years: OR $\approx$ 0.998 (95% CI 0.996-0.999; $p \approx 0.0037$) – a very small negative effect: each additional year of age slightly reduces the odds, conditional on experience and other covariates. Statistically significant, but economically negligible.
- Job level:
 - Entry/Fresher OR $\approx$ 0.78 (0.67-0.90), Junior OR $\approx$ 0.78 (0.68-0.90), Middle OR $\approx$ 0.85 (0.73-0.98), Part-time OR $\approx$ 0.79 (0.67-0.94), Others OR $\approx$ 0.64 (0.51-0.81), null OR $\approx$ 0.48 (0.30-0.76). All have lower odds than the reference.
 - Senior is borderline ($p \approx 0.096$); Top (Directors/VP) is not different from the reference ($p \approx 0.60$).
- Industries:
 - Construction/Cement/Metal/Steel OR $\approx$ 0.85 (0.77-0.95; $p \approx 0.003$) – lower odds.
 - Engineering/Projects OR $\approx$ 1.09 ($p \approx 0.050$) and Entertainment/Media OR $\approx$ 1.11 ($p \approx 0.040$) – higher odds.
 - Most other industries show no material differences.
- Functional areas:
 - Teaching/Education OR $\approx$ 0.84 (0.73-0.96; $p \approx 0.0085$) – lower odds.
 - Most other functional areas are nonsignificant or only marginal.

After controlling for age, industry, and functional area, each year of experience substantially increases the probability of being in the upper-quartile salary group, whereas lower job levels meaningfully reduce it relative to the reference level. The effect of age is small, and industry/functional effects are present but generally weaker than experience. The model is suitable for coarse screening (AUC $\sim$ 0.77), while remaining variance reflects unobserved factors.

Answer to Q3 – “Which factors are associated with a higher likelihood of earning a salary in the top quartile (top 25%)?” – The likelihood of being in the top-25% of salaries is driven primarily by years of experience. Age exhibits a very small, negative association after controlling for experience. Industry and functional area contribute localized effects, but their magnitudes are notably smaller than the effect of experience.

Is the probability of a "high salary" higher in IT-Software compared to other industries, controlling for experience and level?

```
> # Binary flag "IT-Software vs Others"
```

```
> Job$IT_software_flag <- as.integer(Job$Industry == "IT-Software")
> dat2 <- Job %>%
+   dplyr::select(HighPay, Experience_years, Age_years, Level, IT_software_flag)
%>%
+   na.omit()
> m2 <- glm(HighPay ~ Experience_years + Age_years + Level + IT_software_flag,
+           data = dat2, family = binomial())
> summary(m2)
```

Call:

```
glm(formula = HighPay ~ Experience_years + Age_years + Level +
     IT_software_flag, family = binomial(), data = dat2)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-2.1597012	0.0815367	-26.487	< 2e-16

Experience_years	0.1138022	0.0008083	140.794	< 2e-16

Age_years	-0.0024111	0.0008043	-2.998	0.002720
**				
LevelEntry Level/Fresher	-0.2494688	0.0730997	-3.413	0.000643

LevelJunior - Executive, Assistant	-0.2444359	0.0725795	-3.368	0.000758

LevelMiddle - Manager, Assistant Manager	-0.1689081	0.0724040	-2.333	0.019656 *
Levelnull	-0.7712271	0.2359809	-3.268	0.001082
**				
LevelOthers	-0.4530226	0.1161687	-3.900	9.63e-05

LevelPart time	-0.2418674	0.0835516	-2.895	0.003794
**				
LevelSenior -Sr. Manager, General Manager	-0.1225019	0.0737821	-1.660	0.096850 .
LevelTop - Directors, VP	-0.0321494	0.0859993	-0.374	0.708528
IT_software_flag	0.0053140	0.0191963	0.277	0.781914

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 201349 on 178042 degrees of freedom
 Residual deviance: 177504 on 178031 degrees of freedom
 AIC: 177528

Number of Fisher Scoring iterations: 4

> # Odds Ratios (OR) with 95% Confidence Intervals (focus on IT_software_flag)

```
> z <- qnorm(0.975) # 1.96
> or2 <- broom::tidy(m2, conf.int = FALSE) %>%
+ mutate(
+   OR = exp(estimate),
+   OR_lo = exp(estimate - z * std.error), # Wald 95% CI
+   OR_hi = exp(estimate + z * std.error)
+ ) %>%
+ arrange(p.value)
```

> # Leading effects (excluding the intercept)

```
> or2 %>% filter(term != "(Intercept)") %>% head(30)
```

A tibble: 11 × 8

term	estimate	std.error	statistic	
p.value	OR	OR_lo	OR_hi	
<chr>	<dbl>	<dbl>	<dbl>	
<dbl>	<dbl>	<dbl>	<dbl>	
1 Experience_years	0.114	0.000808	141.	0
1.12 1.12 1.12				
2 LevelOthers	-0.453	0.116	-3.90	
0.0000963 0.636 0.506 0.798				
3 LevelEntry Level/Fresher	-0.249	0.0731	-3.41	0.000643
0.779 0.675 0.899				
4 LevelJunior - Executive, Assistant	-0.244	0.0726	-3.37	0.000758
0.783 0.679 0.903				
5 Levelnull	-0.771	0.236	-3.27	0.00108
0.462 0.291 0.734				
6 Age_years	-0.00241	0.000804	-3.00	0.00272
0.998 0.996 0.999				
7 LevelPart time	-0.242	0.0836	-2.89	0.00379
0.785 0.667 0.925				
8 LevelMiddle - Manager, Assistant Manager	-0.169	0.0724	-2.33	0.0197
0.845 0.733 0.973				
9 LevelSenior -Sr. Manager, General Manager	-0.123	0.0738	-1.66	0.0969
0.885 0.766 1.02				
10 LevelTop - Directors, VP	-0.0321	0.0860	-0.374	0.709
0.968 0.818 1.15				

```

11 IT_software_flag          0.00531  0.0192      0.277 0.782
1.01  0.968 1.04
> # Examine IT_software_flag separately
> or2 %>% filter(term == "IT_software_flag") %>%
+   select(term, OR, OR_lo, OR_hi, p.value)
# A tibble: 1 × 5
  term          OR OR_lo OR_hi p.value
  <chr>      <dbl> <dbl> <dbl>   <dbl>
1 IT_software_flag 1.01 0.968 1.04   0.782
> dat1$prob <- predict(m1, type = "response")
> library(pROC)
> roc1 <- pROC::roc(dat1$HighPay, dat1$prob)
Setting levels: control = 0, case = 1
Setting direction: controls < cases

```

We created a binary outcome HighPay as an indicator of being in the top salary quartile (top-25%) based on Salary_lacs. On this basis we assembled dat2 with ~178k observations (after na.omit) containing: HighPay, Experience_years, Age_years, Level, IT_software_flag (1 = industry "IT-Software", 0 = others).

We estimated a logistic regression:

$$\text{logit } P(\text{HighPay}=1) = \beta_0 + \beta_1 \cdot \text{Experience_years} + \beta_2 \cdot \text{Age_years} + \beta_3 \cdot \text{Level} + \beta_4 \cdot \text{IT_software_flag}.$$

IT_software_flag: Estimate = 0.005314, Std. Error = 0.019196, z = 0.277, p = 0.782.

Exponentiating gives $OR = \exp(0.005314) \approx 1.01$.
Wald 95% CI:
lower = $\exp(0.005314 - 1.96 \cdot 0.019196) \approx \exp(-0.0323) \approx 0.968$,
upper = $\exp(0.005314 + 1.96 \cdot 0.019196) \approx \exp(0.0429) \approx 1.04$.
Thus $OR \approx 1.01$ (95% CI: 0.97-1.04; p = 0.782) – statistically non-significant and practically null.

From model m1: when we replace the flag with full industry dummies (and also control for Functional_Area), the coefficient for IndustryIT-Software = -0.0104 with p = 0.809 → $OR \approx 0.99$ (CI includes 1). This independently confirms that belonging to IT-Software does not increase the odds of being in the top-25% once experience and level are controlled for.

Based on model m2 (flag IT vs. others) and consistent with model m1 (detailed industry dummies), membership in IT-Software by itself does not increase the probability of a "high" salary after controlling for experience and job level. The largest and most stable contribution comes from experience (and, to a lesser extent, job level); the effect of the IT-Software industry is null in both statistical and practical terms.

Answer to "Is the probability of a 'high salary' higher in IT-Software, controlling for experience and level?" – No. After adjusting for experience, age, and job level, working in IT-Software is not associated with higher odds of being in the top-25% of salaries. Experience and level are the primary drivers.

```

> thr1 <- as.numeric(pROC::coords(roc1, "best", ret = "threshold"))
> pred1 <- factor(as.integer(dat1$prob >= thr1), levels = c(0,1), labels =
c("No", "Yes"))
> ref1 <- factor(dat1$HighPay, levels = c(0,1), labels =
c("No", "Yes"))
> caret::confusionMatrix(pred1, ref1, positive = "Yes")

```

Confusion Matrix and Statistics

```

      Reference
Prediction  No   Yes
      No  88255 10388
      Yes 44403 34463

      Accuracy : 0.6913
      95% CI : (0.6892, 0.6935)
No Information Rate : 0.7473
P-Value [Acc > NIR] : 1

```

```

      Kappa : 0.3467

```

```

McNemar's Test P-Value : <2e-16

```

```

      Sensitivity : 0.7684
      Specificity : 0.6653
      Pos Pred Value : 0.4370
      Neg Pred Value : 0.8947
      Prevalence : 0.2527
      Detection Rate : 0.1941
      Detection Prevalence : 0.4443
      Balanced Accuracy : 0.7168

```

```

      'Positive' Class : Yes

```

```

> pROC::auc(roc1)

```

Area under the curve: 0.7707

- AUC = 0.7707 The model shows good discrimination (well above random), a threshold-independent measure of ranking quality.
- Confusion matrix (at the chosen threshold):
TP = 34,463; FP = 44,403; TN = 88,255; FN = 10,388 (N = 177,509).
- Sensitivity (Recall) = 0.768 The model correctly identifies ~77% of top-quartile earners.
- Specificity = 0.665 The model correctly rejects ~67% of non-top-quartile cases.
- Positive Predictive Value (Precision) = 0.437 Moderate, which is expected given the relatively low prevalence of the positive class.
- Negative Predictive Value = 0.895 "No" predictions are generally reliable.

- Balanced Accuracy = 0.717 Provides a reasonable compromise between sensitivity and specificity.
- Accuracy = 0.691 is below the No-Information Rate = 0.747; with class imbalance, plain accuracy is not very informative, so we emphasize AUC and recall/precision.

># Experience years significantly increases the chances of HighPay (top-25%); AUC ~ 0.77 (good discriminative ability).

># Some Levels have a substantial impact on the odds (specific coefficients are in the OR table).

># IT-Software has no independent advantage after controlling for experience and level (OR $\approx$ 1; $p \gg 0.05$).

Overall recommendation and findings based on the model

The analysis used the cleaned subset of job-seeker records where compensation is explicitly in Indian Rupees (Rs). Salary was standardized to *lacs per year*; work experience and age were engineered from text/date fields; categorical fields were trimmed to the first tag to avoid explosion of rare levels.

Q1 — Clustering (profiles in age/experience/salary).

Unsupervised k-means ($k \approx 4$) on standardized age, experience, and salary reveals four stable segments rather than a single linear ladder. In words: (i) a large mid-market group with moderate experience and moderate pay; (ii) older candidates with moderate pay; (iii) very experienced candidates who remain on moderate pay; and (iv) a smaller, highly paid senior segment. The key strategic implication is that compensation branches after ~15–20 years of experience: some careers progress into high-pay tracks, while others plateau—so experience alone does not guarantee high pay.

Q2 — Linear regression (what moves salary more?).

Across specifications, years of experience is the most reliable monetary driver of salary ($\approx +0.53$ lacs per additional year, holding age fixed). Age has a small negative association after controlling for experience and is economically negligible. Adding education does not materially improve fit; adding industry introduces local shifts but does not outperform the simpler, more interpretable model. Including job level helps interpret jumps, but a single “step up” in level is typically equivalent to only $\sim \frac{1}{2}$ –1 year of experience in salary terms and is less stable across categories. By AIC, the preferred specification is Model 3: Salary $\sim$ Experience + Age + Level, which balances parsimony and performance better than alternatives.

Q3 — Logistic regression (who is likely to be in the top-25%).

For the binary outcome “HighPay” (top quartile by salary), experience is again the dominant factor: each additional year increases the odds of being in the top quartile by roughly 12% ($OR \approx 1.12$, $p < 0.001$). Age shows a very small negative effect conditional on experience. Most industry and functional-area effects are modest and localized; a few categories differ from baseline, but their magnitudes are notably smaller than the

experience effect. The model exhibits useful discrimination ($AUC \approx 0.77$). At a threshold chosen by Youden's J, recall of top-quartile earners is high with moderate precision—appropriate for a screening use-case where missing strong candidates is costlier than reviewing more profiles.

Practical recommendations.

1. **Prioritize relevant experience.** It is the strongest and most stable predictor of both absolute salary and the probability of being in the top quartile.
2. **Use level judiciously.** Title changes alone deliver smaller, less consistent pay gains than accumulated experience; align promotion bands with demonstrated experience/impact.
3. **Don't over-index on "IT-Software."** After controlling for experience and level, IT-Software candidates are not systematically more likely to be high-paid in this data.
4. **Segment strategy.** Target sourcing and compensation policies by cluster: design distinct value propositions for plateaued senior profiles versus high-trajectory seniors; avoid applying a single linear rule beyond ~15–20 years of experience.
5. **Operationalize the classifier.** Deploy the logistic scores to prioritize reviews. If missing top talent is costly, use a lower threshold (higher recall); if interview bandwidth is limited, raise the threshold to improve precision.

The data are noisy and very heterogeneous, with many rare categories. We're also missing factors like city/region, firm size, and skills, which likely matter. Next, I'd try non-linear terms (e.g., splines for experience) and maybe quantile regression to see how effects change across the pay distribution. Proper train/validation splits and PR curves would help pick thresholds under class imbalance.

I also tested richer specs (more variables, interactions, regularization, stepwise). They didn't beat the chosen models by AIC and took much longer to fit because of the noisy data and sparse categories, even after optimization (parallelization, level pruning, sparse encodings).

If you have to bet on one thing, bet on experience. It consistently explains more about salary levels and who lands in the top pay group. Titles and industries help a bit for interpretation, but they don't replace experience.